Wi-Breath: A WiFi-Based Contactless and Real-Time Respiration Monitoring Scheme for Remote Healthcare

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***Abstract*—Respiration rate is an important healthcare indicator, and it has become a popular research topic in remote healthcare applications with Internet of Things. Existing respiration monitoring systems have limitations in terms of convenience, comfort, and privacy, etc. This paper presents a contactless and real-time respiration monitoring system, the so-called Wi-Breath, based on off-the-shelf WiFi devices. The system monitors respiration with both the amplitude and phase difference of the WiFi channel state information (CSI), which is sensitive to human body micro movement. The phase information of the CSI signal is considered and both the amplitude and phase difference are used. For better respiration detection accuracy, a signal selection method is proposed to select an appropriate signal from the amplitude and phase difference based on a support vector machine (SVM) algorithm. Experimental results demonstrate that the Wi-Breath achieves an accuracy of 91.2% for respiration detection, and has a 17.0% reduction in average error in comparison with state-of-the-art counterparts.**

***Index Terms*—Respiration monitoring, channel state information, signal selection, remote healthcare.**

I. Introduction

BENEFITING from the dramatic progresses of Internet of Things (IoT) paradigms, remote healthcare have rapidly entered the era of digital health such as monitoring, recording, and tracking the key signs of people's health to improve their lifestyle and health disorders [1]–[3]. The respiration is a vital sign for remote and continuous monitoring of the diseases' progression [4]. It has been reported that respiratory rate has similar diagnostic value with that of other physiological measurements such as pulse rate or blood pressure, in discriminating between stable patients and patients at risk [5]. Respiration monitoring has become a hot topic in IoT based E-home monitoring applications [6]–[8].

Current available devices for respiration remote monitoring are based either on contact or contactless technology. Generally, the contact methods are more accurate than the contactless alternatives. They can be divided into six categories, acoustic-based methods, airflow-based methods, chest and abdominal movement methods, transcutaneous CO_2 measurement methods, oximetric (SpO_2) methods and electrocardiographic (ECG) methods [4]. The greater accuracy of the contact devices provides advantages in short-term clinical detection leading to better diagnosis and treatment. However, contact devices are uncomfortable and inconvenient for users [9]–[11]. Improved comfort and convenience for patients have been the drivers for the development of contactless detection devices. Video-based systems detect respiration by monitoring the movement of the chest [12]–[14]. However, they do not work well in weak-light conditions and have user privacy concerns. Systems based on infrared cameras avoid the weak-light problem, but the required high resolution cameras are expensive [15]. Radio frequency (RF) techniques based on Doppler radar signals (2.4 GHz) have been proposed for this purpose [16], but unlike the device proposed here, they require interfacing to additional expensive antenna equipment. Due to the limitations mentioned above, few contactless systems have been applied in long-term monitoring.

Recently, the rapid development of WiFi has stimulated research into the use of off-the-shelf WiFi devices for monitoring human activities in a domestic setting [17]–[19]. The main advantages of WiFi based respiration monitoring systems are their contactless nature, making them more comfortable and less cumbersome, as well as their low cost. Received signal strength (RSS) is generally used as a selection criterion in existing WiFi based respiration detection systems [20]. However, the RSS

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data only provides an overview of the channel, but does not yield information about the various subcarriers. Furthermore, the current systems based on RSS have low resolution in time and amplitude [21]–[23].

The direction, amplitude, polarization and phase of the signal are changed by the reflection from objects in the beam path, giving rise to multipath effects. Channel state information (CSI) is another descriptor of WiFi signals, from which the information about the multipath effects and different subcarriers can be obtained and measured by orthogonal frequency division multiplexing (OFDM) rather than relying on the overall channel frequency response (CFR) [24]. These subcarriers, by providing multiple channels of data are more effective for respiration monitoring. By exploiting fine-grained channel information, many groups have proposed monitoring systems based on CSI signals. The amplitude component of CSI is widely used in the detection of respiratory rate, achieving better performance than RSS based systems [25]–[27]. However, phase information is also available in CSI and, furthermore, when two antennas are used, the phase difference between them is sensitive to micro-movements, and can thus be used for detecting activity [28]. In this paper, a respiration monitoring system (Wi-Breath) based on both amplitude and phase-difference information of the CSI, is developed. This system differs from existing ones by providing three contributions:

1) The phase information of the CSI signal is considered and both the amplitude and phase difference are used.
2) For better respiration detection accuracy, a signal selection method is proposed to select an appropriate signal from the amplitude and phase difference of the CSI, based on a SVM algorithm.
3) A real-time respiration detection system, Wi-Breath, is implemented based on WiFi devices, and validation experiments show that the accuracy of Wi-Breath is better than that of existing respiration detection methods based on CSI amplitude signals.

The remainder of this paper is organized as follows. A brief review of previous work on respiration monitoring is presented in Section II. Then, Section III introduces the CSI model of WiFi for respiration detection. The Wi-Breath system is presented in Section IV, which includes signal pre-processing, signal selection and respiration rate calculation. Section V presents the results of system performance, and finally, conclusions are drawn in Section VI.

II. Related Work

Generally, respiration detection systems can provide accurate results by using contact sensors. For example, a thermistor can be used to detect respiration by monitoring the change of temperature due to airflow variations in the nose [9]. Straps measuring chest and abdominal movement during respiration have been widely used in hospitals, and these give accurate respiration rates [10], [11]. In a transcutaneous CO_2 monitoring method, a heated electrode (about 42 °C) is applied to monitor variations in the diffusion of gas from blood to the skin and to obtain an overall estimate of changes in CO_2 level and thus to detect time-averaged consequences of abnormal respiration, rather than beat-to-beat variation [29].

TABLE I
Recent Work for Respiration Monitoring Compared With the Wi-Breath System

	System advantages			Signal information	
Methods	Off-the-shelf device	Privacy protection	Contact-less device	Multi-Path	Phase info.
SpO_2/CO_2 [4], [29]		✓		-	-
ECG [4]		✓		-	-
Airflow [4]		✓		-	-
Straps [10], [11]		✓		-	-
Camera [12]	✓		✓	-	-
Doppler radar [16]		✓	✓		✓
RSS [20], [21]	✓	✓	✓		
CSI Amp [25]–[27]	✓	✓	✓	✓	
Wi-Breath	✓	✓	✓	✓	✓

✓(Yes); × (No); - (Not applicable).

As a contactless monitoring device, a non-restrictive optical sensing method for respiration monitoring with a fiber grating (FG) sensor and processor unit has been reported [30]. A camera has been used to detect thoracic movements to measure respiratory rate, but this inevitably imposes on the subject's privacy [12], [13]. In addition, RF devices have been used to analyze physiological movement [16], [31]. For example, a 2.4 GHz antenna array was designed to monitor vital signs, including respiration and heart rate [16], but the users need to buy additional expensive antenna devices.

On the other hand, WiFi-based respiration monitoring systems can use off-the-shelf devices. In many such systems, RSS signals have been used to detect respiration and thus measure its rate. For example, UbiBreath [20], a WiFi-based respiration monitoring system, detects RSS signals, but its performance is influenced by changes in the subjects' posture and location of the equipment. BreatheTaking [21] uses a maximum likelihood estimator (MLE) for deriving respiration rate and takes measurements of RSS between many pairs of wireless devices in a deployed network. In spite of this, systems based on RSS lack multipath information and this impairs their performance.

To address these issues, CSI has been widely used to monitor vital signs including respiration signals. Compared to RSS, it can acquire more channel information including the amplitude and phase of each subcarrier. The amplitude of CSI signals has been widely used in measuring body micro movement. 90% of the respiration measurements based on the amplitude of CSI give an error of less than 1 breath per minute (bpm) [25]. Some respiration monitoring methods based on the amplitude of CSI signals take into account the influence of different postures and also variations in the distance between the subject and the device, as well as variations in their relative orientation, and these perform better than those based on the RSS technique [26], [27].

Table I is a comparison of the Wi-Breath system with recent work on respiration monitoring. To summarize, the Wi-Breath

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system can detect respiration using contactless low cost off-the-shelf devices, which minimize users' discomfort and protect their privacy. Besides, it uses the CSI signals which have the multi-path information. The phase information of CSI is also introduced, which can provide more useful information to improve the accuracy of respiration detection.

III. Channel State Information Model for Tracking Respiration

A. Channel State Information

Multipath propagation introduces selective fading in the frequency domain. CFR obtained from the CSI is used to characterize multipath propagation including amplitude and phase in each subcarrier, which supports IEEE 802.11a/g/n. CSI, which in a typical off-the shelf device has 30 subcarriers in each channel, thus contains a lot more channel information than RSS. The amplitude and phase of each subcarrier can be described by

$$\hat{H}(f_k) = ||\hat{H}(f_k)||e^{j\hat{\phi}_{f_k}}, \tag{1}$$

where $\hat{H}(f_k)$ is the CSI of the kth subcarrier with measured frequency f_k, and $||\hat{H}(f_k)||$ and $\hat{\phi}_{f_k}$ denote its measured amplitude and phase.

The measured amplitude $||\hat{H}(f_k)||$ of CSI in one subcarrier always contains the theoretical value $||H(f_k)||$ and the environment noise is

$$||\hat{H}(f_k)|| = ||H(f_k)|| + Z_A, \tag{2}$$

where Z_A is the environment noise in amplitude. In addition, the measured phase of CSI $\hat{\phi}_{f_k}$ [32] can be computed in each subcarrier as

$$\hat{\phi}_{f_k} = \phi_{f_k} + 2\pi f_k \Delta t + \beta + Z_P, \tag{3}$$

where ϕ_{f_k} is the theoretical phase, Δt is the time lag at the same antenna, β is an unknown constant phase offset, Z_P is environmental noise in phase.

B. CSI Signal for Respiration Monitoring

Subject micro movement can affect the propagation of a WiFi signal, which will cause the changes in the amplitude and phase of the CSI signals. Respiration is an example of this kind of micro movement and can thus also be detected by CSI. The variation of the amplitude and phase signals of CSI caused by respiration are shown in Fig. 1(a) and Fig. 1(b). The amplitude signal clearly shows variations at a rate consistent with respiration. On the other hand, there are no easily recognizable respiratory features in the phase signal. In existing devices f_k is in the region of 5 GHz which, from Eq. (3), means that a small change in time lag Δt can cause a large phase offset. As a result, the original phase does not have the typical characteristics of breathing when monitoring respiration.

However, if we consider a two-antenna system and calculate the phase difference according to Eq. (3), where the value of f_k in each antenna is the same (5 GHz), the measured phase difference $\Delta\hat{\phi}_{f_k}$ is shown as

$$\Delta\hat{\phi}_{f_k} = \Delta\phi_{f_k} + 2\pi f_k(\Delta t_1 - \Delta t_2) + \Delta\beta + \Delta Z_P, \tag{4}$$

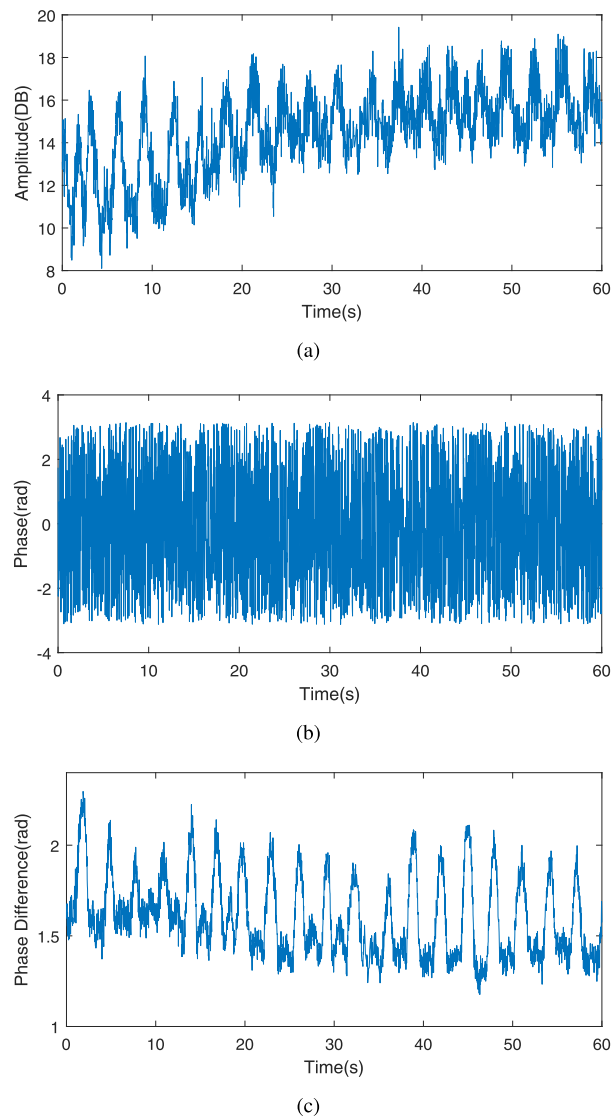

Fig. 1. Amplitude, original phase and phase difference of the CSI signal in the respiration wave: (a) the original amplitude signal; (b) the original phase signal; (c) the phase difference signal.

where $\Delta\phi_{f_k}$ is the theoretical phase difference between two signals, which can reflect the change of the transmission path. Δt_1 is the time lag at the first antenna, Δt_2 is the time lag at the second antenna, $\Delta\beta$ is the constant phase offset and ΔZ_P is the environmental noise difference between the two antennas. When the CSI signals from two antennas are measured at the same wireless device, Δt_1 is approximately equal to Δt_2 [29]. Thus the phase difference reduces to

$$\Delta\hat{\phi}_{f_k} = \Delta\phi_{f_k} + \Delta\beta + \Delta Z_P. \tag{5}$$

When the environment is stable, the theoretical phase difference $\Delta\phi_{f_k}$ between two antennas is almost a constant. When micro movement of human body occurs, the phase difference $\Delta\phi_{f_k}$ will change because of the change in the transmission path. As a result, the measured phase difference $\Delta\hat{\phi}_{f_k}$ will change according to (5). Therefore, the measured phase difference can

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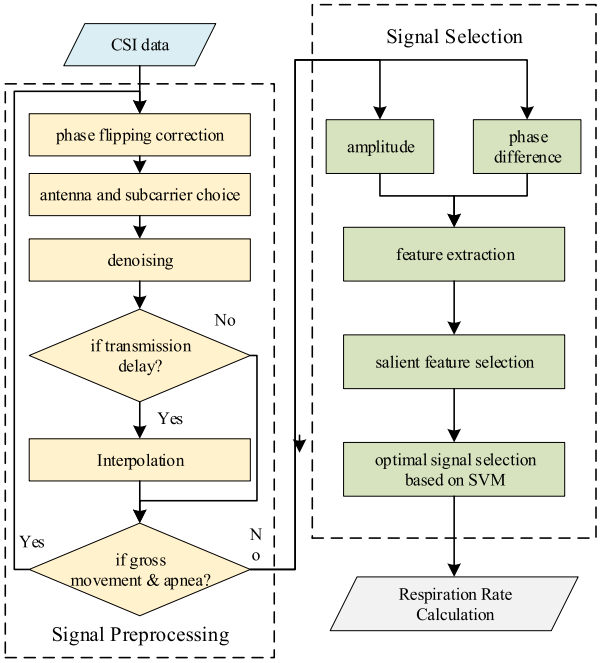

Fig. 2. The flow diagram of the Wi-Breath system.

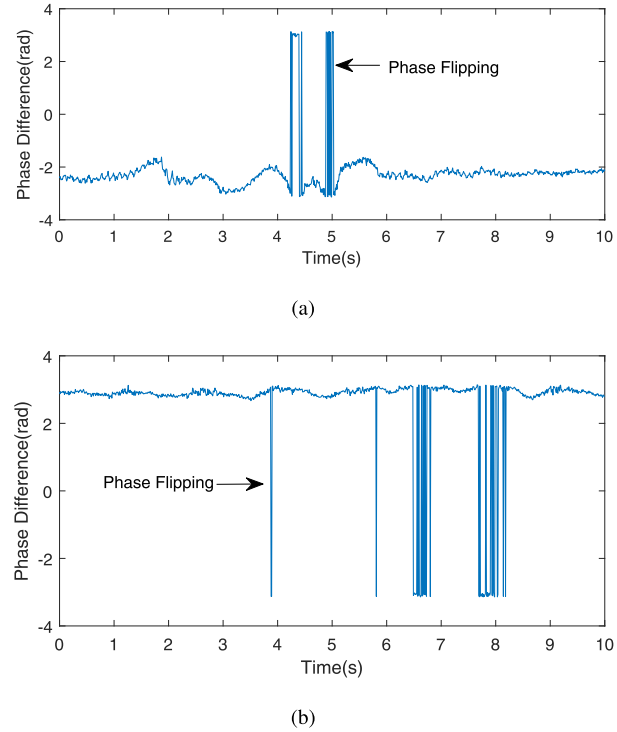

Fig. 3. Phase flipping of phase difference signal: (a) phase flipping less than -π; (b) phase flipping more than π.

also be used to detect micro movement including respiration. Fig. 1(c) shows the phase difference of the CSI signals obtained from two antennas. It illustrates that the phase difference has the same regularity as the amplitude wave (Fig. 1(a)). Thus, both amplitude and phase difference signals can be used for respiration rate monitoring.

IV. System and Method

A. System Overview

In the Wi-Breath system, a wireless card with 3 antennas and a router with 2 antennas were used to acquire the CSI signals. The wireless signal frequency is 5 GHz. As mentioned, compared with other WiFi-based monitoring systems, Wi-Breath uses both amplitude and phase difference signals to obtain additional respiration information. As illustrated in Fig. 2, the software can be divided into two parts: (1) signal pre-processing, (2) signal selection and respiratory rate calculation.

1) Signal Pre-Processing: The pre-processing segment analyzes the signal quality. First of all, the phase flipping of phase difference signals is detected and corrected (see below). Then, the optimal antenna and subcarrier are selected. Next, the environmental interference is denoised to enhance the signal. Then, the sample rate is determined and the transmission delay is corrected by linear interpolation in real-time. Finally, aberrant signals including those due to gross movement and apnea are processed.

2) Signal Selection and Respiratory Rate Calculation: In this segment, we select the optimal signal from the amplitude and phase difference signals using the SVM classifier to calculate the respiration rate. First, we extract the respiration features from the two signals. Then the salient features are determined by a backward elimination algorithm. Finally, we build the appropriate signal classification model based on the SVM algorithm to select the optimum signal for respiration rate calculation.

B. Signal Pre-Processing

1) Phase Flipping Correction: In the CSI signals, there are some outliers caused by phase flipping. According to (1), the CSI signal is represented as a complex function and the phase signal of CSI is constrained to the range of -π to π. When the phase is more than π or less than $-\pi$, the phase will be flipped due to the limited range shown in Fig. 3(a) and Fig. 3(b). Clearly, the phase flipping can be detected if an absolute difference in the order of 2π between adjacent points of the phase difference signal is seen. Once detected, it can be corrected by subtracting or adding 2π from current value of the flipped phase difference signal.

2) Choice of Antenna and Subcarrier: There are 2 transmitting antennas (TXs) and 3 receiving antennas (RXs) in our system. Each TX and RX pair forms a transmit link, so there are 6 transmit links in the system. In addition, there are 30 subcarriers received in the channel of each transmit link. Thus, when the system acquires the CSI information from each received packet, we extract a $2 \times 3 \times 30$ matrix (RX $\times$ TX$\times$ CFR) from the six possible RX-TX antenna pairs, and obtain a total of 12 average amplitude and phase signals, which are averaged out from all 30 individual subcarrier signals. There is much variation between the 6 channels, so it is necessary to identify the "best" RX-TX antenna pair. This was done by selecting the channel with the greatest signal to noise ratio. The signal to noise ratio (SNR) for

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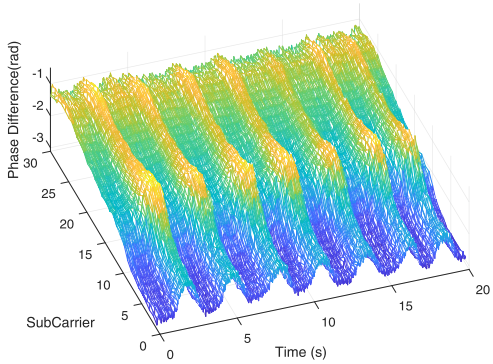

Fig. 4. Different 30 subcarriers in phase difference signal in one channel.

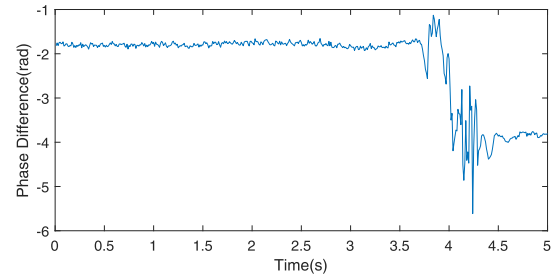

Fig. 5. The phase difference signal due to gross movement.

the i-th RX-TX antenna pair, ANTs can be calculated by

$$SNR(ANTs) = \max\{10 \times \log_{10}(Ps_i\ /Pn_i\ )\}, i = (1, .., 6), \quad (6)$$

where Ps_i and Pn_i are the average power levels of the signal and noise of the 30 subcarriers in one RX-TX antenna pair.

Fig. 4 shows that the phase difference signals of the 30 different subcarriers in one RX-TX antenna pair also have a wide range of SNR levels. This implies that the performance of the sytem can be optimized by choosing the best subcarrier in each RX-TX pair.

We propose an algorithm to select the best subcarrier. Generally, we need to calculate the SNR to evaluate the signal quality. However, this is a time consuming process and is therefore not feasible for real-time monitoring. We assume that environmental noise is stable for a period of time, and the max-min amplitude range of the sub-carriers can be regarded as a standard instead of SNR. It can be expressed by

$$quar(CFR_s) = \max\{quar(CFR_i)\}, i = \{1, \ldots, 30\}, \quad (7)$$

where CFR_s is the best subcarrier which is selected, CFR_i is the ith subcarrier and *quar* is the inter quartile range.

C. Denoising

In Eq. (2) and (5), the environmental noise which adds to the respiration signal is denoted by Z_A and ΔZ_f. We have attempted to reduce this by bandpass filtering (e.g. Butterworth). However, this will affect the detection accuracy of outlier signals, such as those due to apnea and gross movement. To minimize this problem, we apply the discrete wavelet transform (DWT) for denoising. To get an efficient wavelet basis, we compared the effect of several different basis functions based on SNR, and finally, chose 'db4' to filter the CSI signals.

D. Sample Rate Selection and Transmission Delay Correction

Sample rate can influence the detection accuracy of respiration. A low sample rate imposes a low time resolution on the CSI signal. On the other hand, package losses can occur in signal transmission at high sample rates. The sample rate selection is discussed in Section V-C. Additionally, delays in signal transmission caused by environmental interference will make the signal transmission interval longer than the pre-set value of the sample rate. This will lead to signal distortion and errors in calculating the respiration rate. To reduce the effect of transmission delay on signal sampling, a delay detection and correction algorithm is proposed. When the package transmission time t is longer than the sample interval T, a transmission delay is detected. The Wi-Breath system notes the duration of the delay and adds points by linear interpolation between adjacent real sample points in the received signal. This can be described by Eq. (8), where λ is the number of points interpolated, t is package transmission time and T is sample interval, and *Int()* denotes an integral function.

$$\lambda = \begin{cases} 0, \text{ if } t < T \\ Int(t/T), \text{ if } t > T \end{cases}. \quad (8)$$

E. Aberrant Signal Detection

There are two kinds of aberrant signals considered in the Wi-Breath system: those due to gross movement and apnea. These can be detected and processed as follows.

1) Gross Movement Detection: This occurs when users move their bodies, causing a sudden change in the CSI signal, as shown in Fig. 5. Thus, gross movement can be detected in a 10 seconds sliding window moving once per second by a threshold (2 rads) of the change in phase difference, which has been determined empirically. When the gross movement occurs, the respiration rate calculation is paused until the signal is deemed to be free of movement artefact.

2) Apnea Detection: Apnea is characterised by pauses in breathing and will thus affect the measured respiration rate [33]. It is often associated with obesity [34], high blood pressure, myocardial infarction, myocardial ischaemia and stroke [35]. As illustrated in Fig. 6, movement due to inhalation may cease for about 10 seconds during apnea. This may occur up to 20-30 times each night and can lead to a generalised impairment of tissue oxygenation and consequent medical complications. Thus, it is important to detect apnea early to minimise its adverse effects. To detect apnea, a variable, θ, is calculated as the ratio of two moving averages over a 10 s time window by

$$\theta = P_{(t-10,t)}/P_{(t,t+10)}, \quad (9)$$

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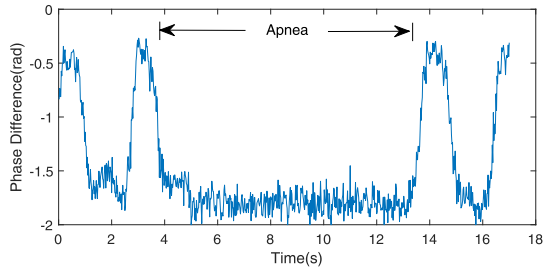

Fig. 6. The phase difference signal wave during apnea.

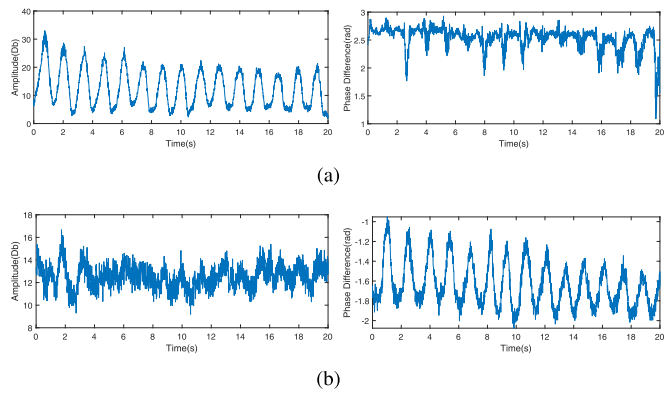

Fig. 7. Phase flipping of phase difference signal: (a) amplitude signal quality higher than that of the phase difference; (b) signal quality of the phase difference is higher than that of the amplitude.

where t is the current measurement time, $P_{(t-10,t)}$ is the average power of the signal during the preceding 10 seconds and $P_{(t,t+10)}$ is the power of signal during the subsequent 10 seconds. Both the two sliding windows move once per second. If apnea has occurred in the previous 10 seconds, θ will be below 20% of its value during normal breathing [36], and the respiration rate calculation is deferred until θ has returned to an acceptable value.

3) Signal Selection and Respiration Rate Calculation: After pre-processing, the amplitude and phase difference signals were considered suitable for deriving respiration rate. Often the amplitude and phase difference signals are of disparate quality, as shown in Fig. 7. Thus it is necessary to choose which of the two is the most appropriate for respiratory rate calculation. The SVM algorithm is applied as a classification problem to select the optimal signal. Firstly, selected features of two signals are extracted. Secondly, salient features are selected by a backward elimination algorithm. Thirdly, the appropriate signal selection model is trained by using SVM and the optimal signal is determined by the model. Finally, the respiration rate is calculated using the selected signal.

Feature Extraction: To select a signal suitable for further processing, 18 features are extracted, 8 each from the amplitude and phase difference signals and two additional features based on a combination of the signals. These are: 1) SNR of amplitude signal 2) SNR of phase difference signal, 3) amplitude range of original signal, 4) phase difference range of original signal, 5) amplitude range of filtered signal, 6) phase difference range of filtered signal, 7) standard deviation (SD) of amplitude signal, 8) SD of phase difference signal, 9) absolute value of amplitude information entropy, 10) absolute value of phase difference information entropy, 11) SD of respiration rate based on amplitude signal, 12) SD of respiration rate based on phase difference signal, 13) maximum value of respiration rate based on amplitude signal, 14) maximum value of respiration rate based on phase difference signal, 15) average value of respiration rate based on amplitude signal, 16) average value of respiration rate based on phase difference signal, 17) division between the phase difference SD and amplitude SD, 18) division between phase difference SNR and amplitude SNR.

TABLE II
SEVEN MOST SALIENT FEATURES DERIVED FROM THE AMPLITUDE AND PHASE DIFFERENCE SIGNALS

Name	Signal	Salient features	Symbolised as
A_1	*Amp.*	5)Amplitude range of filtered signal	A_1=Range (*Amp.*)
A_2	*Amp.*	7) Standard deviation of amplitude signal	A_2=SD (*Amp.*)
A_3	*Amp.*	15) Average value of respiration rate based on amplitude signal	A_3=Mean (*Amp.*)
P_1	*Phase Diff.*	6) phase difference range of filtered signal	P_1=Range (*Phase Diff.*)
P_2	*Phase Diff.*	4) phase difference range of original signal	P_2= Range (*Original Phase Diff.*)
P_3	*Phase Diff.*	8) Standard deviation of phase difference	P_3=SD (*Phase Diff.*)
C_1	*Amp. & Phase Diff.*	17) division between the phase difference SD and amplitude SD	C_1=A_2/P_3

Feature Selection: To improve the efficiency and accuracy of signal selection, we extracted the most salient features from the 18 listed above by the backward elimination algorithm which filters all features and removes them iteratively one by one [37], [38]. This process resulted in, the 7 salient features, (3 each from the amplitude and phase difference signals and 1 from a combination of the two) listed in Table II.

For the training model, the feature matrix X including the 7 salient features and labels is defined as

$$X = [L, x_{A1}, x_{A2}, x_{A3}, x_{P1}, x_{P2}, x_{P3}, x_{C1}] = \begin{pmatrix} L(1) & x_{A1}(1) & \dots & x_{C1}(1) \\ L(2) & x_{A1}(2) & \dots & x_{C1}(2) \\ \dots & \dots & \dots & \dots \\ L(n) & x_{A1}(n) & \dots & x_{C1}(n) \end{pmatrix}, \quad (10)$$

where L is the label of each training signal, x_{A1} to x_{C1} are the seven selected salient features of the amplitude and phase difference signals and n is the number of training signals.

Signal Selection: All the CSI training data, including amplitude and phase difference signals, were used to calculate respiration rates. To evaluate the performance of all training

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data for respiration detection, a respiration band measurement method was used as the reference standard [20], and then all training data were divided into two categories and labelled accordingly. These were a), signals with higher accuracy of respiration rate calculated based on amplitude, and b) signals with higher accuracy of respiration rate calculated based on phase difference. The optimal signal selection model for respiration detection is trained by SVM based on the labeled features and the optimal signal is determined by using the predicted label which is calculated by trained model. In addition, the kernels of SVM affect the accuracy of signal selection. We compared linear and polynomial kernels and radial basis function. We found that the linear kernel has higher efficiency in signal selection, and that it avoids overfitting, and therefore selected this for use in our system.

Respiration Rate Calculation: When the optimal signal is determined, the respiration rate in breaths per minute (bpm) can be calculated. For better time resolution and to reflect the real-time variability of respiration, we use respiration interval instead of the average rate over a minute, often used in studies of respiration abnormalities [25]–[27]. Here, the respiration interval $T_{interval}$ and the instantaneous respiration rate $R_{instant}$ can be calculated by

$$T_{interval} = T_j - T_{j-1}, \tag{11}$$

$$R_{instant} = 60/T_{interval}, \tag{12}$$

where T_j is the time corresponding to the jth peak of CSI signal wave caused by respiration, and T_{j-1} is the time corresponding to the (j-1)th peak. A sliding window of about 5 seconds in duration is used to detect the peak of CSI signal wave caused by respiration, since normal respiration rate is more than 12 times per minute [39].

V. Experiments and Results

A. Experiment Set-Up and Evaluations

In the Wi-Breath system, a wireless card (Intel 5300) in a personal computer (Lenovo M6900) was used as the receiver, and an 802.11 AP (Mi Router 3), as the transmitter. There are 3 receiver antennas on the wireless card, and 2 transmitter antennas on the router. For better signal quality and to allow more flexibility in placing the antennas, we extended the receiver antennas with 3 SMA extension lines.

Five subjects participated in the experiments. The distance between the transmitter and receiver was 2.5 m. The subjects lay supine in the line-of-sight (LOS) between the RX and TX, and the personal computer analysed and recorded the amplitude and phase difference signals to calculate the respiration rate.

The performance of the Wi-Breath system for respiration detection was evaluated by comparing it to the respiration band method. The average accuracy (*Acc*) and average error (*Err*) of the respiration interval in one test were calculated by

$$Acc = AVR\left(\frac{t_{std} - |t_{std} - t_{CSI}|}{t_{std}}\right), \tag{13}$$

$$Err = AVR(|t_{std} - t_{CSI}|), \tag{14}$$

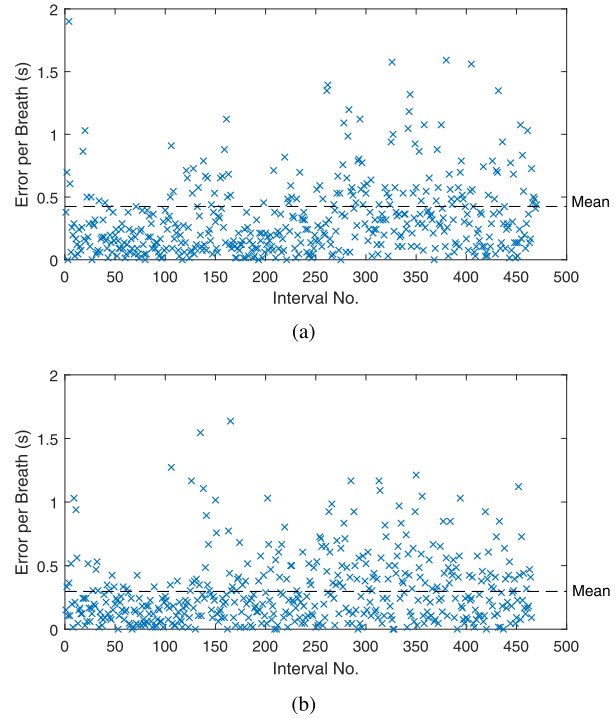

Fig. 8. The error scatter diagrams of respiration interval: (a) error of each respiration interval in the amplitude-based system; (b) error of each respiration interval in the Wi-Breath system.

where t_{std} is the respiration interval of the respiration band, t_{CSI} is the respiration interval of the Wi-Breath system and *AVR* is the value averaged over one minute.

B. Accuracy Evaluation

In this experiment, 200 groups of CSI signals from 5 subjects (30 seconds per group, sampled at different times) were used as a training dataset. 7 salient features from the 18 recorded were extracted with the backward elimination algorithm, and the optimal signal model was trained with the SVM algorithm. With repeated 10-fold cross validation, the average *Acc* of the Wi-Breath in respiration detection reached 91.2%. We also compared the respiration interval errors of the Wi-Breath system with those from amplitude-based systems reported previously [25], [26]. Fig. 8 shows respiration interval error scatter diagrams of an amplitude-based system (a) and the Wi-Breath system (b). The plots are based on a continuous 30 minute recording of CSI signal, with the respiration band method taken as the standard. In the Wi-Breath system, the percentage of errors greater than 0.5 s was 18.5%, against 34.2% for the amplitude-based method.

Furthermore, we compared the average *Acc* and average *Err* of the respiration intervals obtained in four different ways based on the 200 groups of CSI signals. These were, firstly, manually selected optimal amplitude and phase difference signals (as the reference method). Secondly, the respiration rate obtained from the amplitude signals alone (as reported by others). Thirdly, the method based on phase difference signals alone, and finally, the method based on the Wi-Breath system, in which respiration

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TABLE III
COMPARISON OF AVERAGE ACC AND AVERAGE ERR BETWEEN THE WI-BREATH SYSTEM AND 3 OTHER METHODS

	Wi-Breath	Phase Diff.	Amplitude	Manual
AVR(Err)	0.2924s	0.3370s	0.3523s	0.2867s
AVR(Acc)	91.2%	89.9%	89.4%	91.4%
Max(Acc)	99.7%	99.7%	99.7%	99.7%
Min(Acc)	63.4%	63.5%	52.9%	63.5%

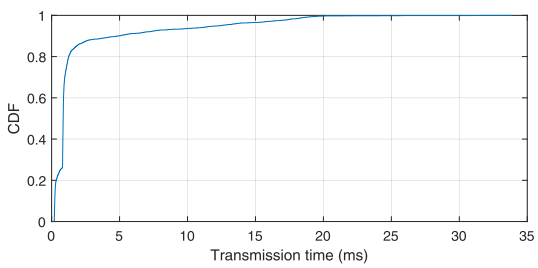

Fig. 9. The CDF of CSI signal transmission time.

TABLE IV
EFFECT OF SAMPLE RATE ON ACCURACY OF DERIVED RESPIRATION RATE

	10Hz	20Hz	50Hz	100Hz
AVR(Acc)	89.7%	89.3%	91.2%	87.2%
Max(Acc)	95.2%	94.6%	97.9%	92.0%
Min(Acc)	87.4%	86.8%	88.7%	83.5%

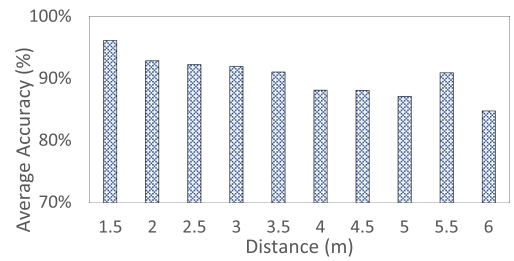

Fig. 10. The effect of RX and TX separation on the accuracy of measured respiration rate.

rate is derived from the combination of signal amplitude and phase difference. Table III compares the performance of these 4 methods. The Wi-Breath system has a lower average *Err* by 0.0599 s and a 17.0% reduction when compared to the amplitude method. Against the phase method, the Wi-Breath system had a 13.2% reduction in average *Err*. The results show that the method based on the phase difference signal performs better than the amplitude signal, and the Wi-Breath system has higher average *Acc* than both of them. In addition, the Wi-Breath system has only a 0.0057 s greater average *Err* and 0.2% less average *Acc* than the average *Acc* of the manual selection method. It shows that the automatic signal selection method of the Wi-Breath system performs well, with an average *Acc* close to the reference method, and that the Wi-Breath system is more accurate than the amplitude-based and phase difference-based methods.

C. Impact of Sample Rate and Transmission Delay Correction

CSI signal transmission delay and package loss will influence the accuracy of respiration rate measurement. Fig. 9 shows the cumulative distribution function (CDF) of the CSI signal transmission time in the Wi-Breath system, based on 20 groups of signals recorded in one minute. It can be seen that the CDF approaches 1 when the transmission time per package reaches 20 ms, suggesting that 50 Hz is an adequate sample rate. For transmission times smaller than 20 ms, the CDF less than 1, which implies that for sample rates greater than 50 Hz, package losses in signal transmission will occur.

To confirm this we investigated the effect of sample rate on package loss. Ten groups of CSI signal were sampled over 30 seconds at rates of 10, 20, 50 and 100 Hz. Based on these signals, the average *Acc* for the different sample rate are shown in Table IV. For sample rates of 10 20 and 50 Hz, the average *Acc* increases from 89.7% to 91% and then falls to 87.2% at 100 Hz. The highest and the lowest Acc values at 50 Hz are also higher than those from the other sample rates, showing that this sample rate yields the best performance. This confirms the notion that for the lower sample rates, the timing has too low a resolution; and at 100 Hz, the package loss increases, leading to reduced accuracy. Nevertheless, for a 50 Hz sample rate there will be occasions when the transmission time exceeds 20 ms and these should be allowed for. We have interpolated data in the received signal for this purpose, according to Eq. (8). To evaluate this delay correction method we used 20 groups of CSI signals recorded over 30 seconds in which there are a few transmission time intervals longer than 20 ms. When the delay correction was applied, the average *Err* decreased from 0.30 s to 0.28 s, a 6.7% improvement.

D. Effect of Tx and Rx Separation

We investigated the effect of the LOS length on the performance of the Wi-Breath system by varying the distance between the RX and TX from 1.5 to 6.5 m in 0.5 m increments. At each distance 10 groups of CSI signals were recorded for 30 seconds. Fig. 10 shows the average *Acc* at each distance. The average *Acc* changed little between separations of 1.5 to 2.5 m, (91-92%) and thereafter, showed a downward trend, due probably to the decreasing received signal strength. Given the typical size of a large bed, we suggest 2.5 m as a workable separation between the transmitter and wireless card.

E. Influence of Posture

Respiration rate is an important index of sleep quality. Sleeping posture is commonly divided into three types [20], supine (soldier and starfish), free-faller and sideways (foetus, log and yearner), as shown in Fig. 11. To investigate the relationship between the accuracy of respiration measurement and posture, 25 groups CSI signals for each posture were used to calculate the respiration rate. These signals came from 5 subjects, and every

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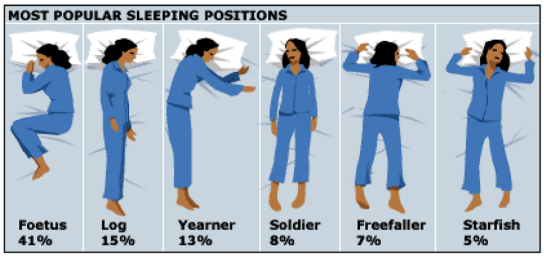

Fig. 11. Six common postures in sleeping [25].

TABLE V
EFFECT OF POSTURE ON ACCURACY OF RESPIRATION RATE MEASUREMENT

	Supine	Freefaller	Sideways
AVR(Acc)	92.8%	90.2%	89.3%
Max(Acc)	95.2%	92.5%	91.7%
Min(Acc)	90.1%	89.3%	88.7%

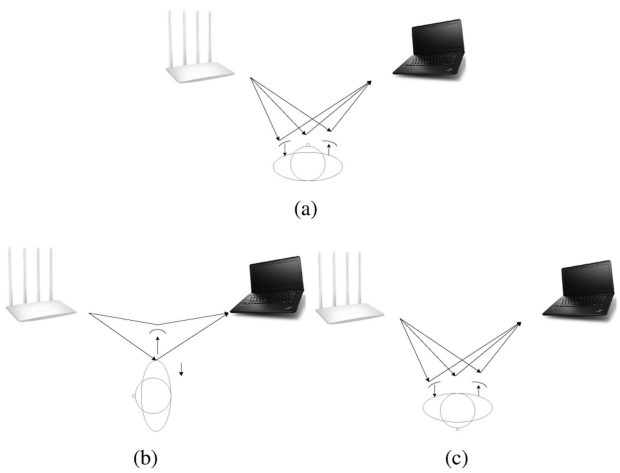

Fig. 12. Schematic diagrams of the different reflection areas and fluctuation range in 3 types of postures: (a) supine; (b) sideways; (c) freefaller.

signal was recorded for 30 seconds. The results are shown in Table V. The supine postures (soldier and starfish) yielded the highest accuracy (average *Acc* 92.8%). The freefaller posture was marginally less accurate at 90.2%; and Sideways (log and yearner), slightly less at 89.3%.

The observed differences in the accuracy of the respiration measurements are presumably due to the differences in the reflection charateristics of the WiFi signal associated with each type of postures, as shown in Fig. 12. It can be seen that the supine posture presents a larger reflection area for the WiFi signal, while the sideways posture presents a smaller reflection area. In the freefaller position, the variation of the WiFi signal during breathing is less than in the supine position, since the back moves less than the chest during respiration [40].

TABLE VI
AVERAGE ACC OF RESPIRATION DETECTION WITH DIFFERENT OBSTACLES

Obstacle	None	Wooden board	Concrete wall
AVR(Acc)	91.2%	88.7%	87.0%
Max(Acc)	93.2%	91.6%	91.9%
Min(Acc)	88.1%	85.8%	82.3%

F. Effect of Obstacles in the Beam Path

Obstacles such as doors, walls and furniture will attenuate the WiFi signals in various ways and thus have an impact on the accuracy of respiration detection. To evaluate the performance of the Wi-Breath system in the presence of obstacles, a wooden board of thickness 23 mm, width 1 m and length 2 m or a concrete barrier of thickness 10 cm, width 5 m and length 4 m were positioned between the router and wireless card in the LOS, and subjects were positioned closed to wireless card by adopting a supine posture. 15 groups of CSI signals recorded for 30 seconds from each of the 5 subjects in the presence of each obstacle and with no obstruction, were analyzed. The distance between RX and TX was 2.5 meters. Table VI shows the average, maximum and minimum *Acc* of respiration detection. With no obstacle in the beam path, the average *Acc* was 91.2%. With the wooden board near the user, the average *Acc* fell to 88.7%; and with the concrete barrier, the average *Acc* was further reduced to 87.0%. This suggests that the system works with acceptable accuracy in the presence of obstacles.

VI. CONCLUSION

In this paper, a respiration monitoring system, Wi-Breath, is developed based on both amplitude and phase difference of the CSI signals. For higher detection accuracy, a signal selection method is proposed using a SVM algorithm, which can determine the optimum signals from the amplitude and phase difference of the CSI to calculate the respiration rate. The experimental results show that the Wi-Breath system can monitor breathing in real-time and has higher accuracy than other respiration detection methods based on WiFi signals. However, there are some limitations in the Wi-Breath system. Data were obtained from only a few volunteers. Thus, the data set is not large enough to avoid overfitting. In addition, to avoid mutual interference, the system can currently only measure one person's respiration. Our future work will address these issues.

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